

Move a condition's border in the health record, and ask how far the genetics move

One worked example with real numbers: sleep apnoea in FinnGen R12 and depression

■ first definition ■ alternative definitions ■ noise, 1.96 SE of the difference

1 Three borders

sleep apnoea, FinnGen R12
cases per definition

hospital records

56,885

+ primary care

62,722

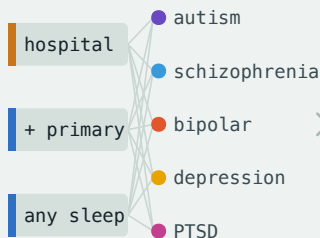
any sleep disorder

65,076

dashed: hospital length

2 Five GWAS each

LD score regression
28 endpoints in all



3 The rg moves

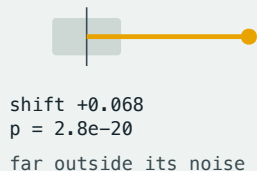
genetic correlation
with depression



4 Beyond its noise?

shared-block jackknife
difference test

noise: 1.96 SE = 0.014



CHECKS BESIDE THE DESIGN

○ published autism heritability and ADHD rg reproduced
○ definition codes mapped to SNOMED CT through OMOP

○ positive control: EHR against clinical depression
○ Finnish against European LD scores

READING THIS

Widening sleep apnoea to any sleep disorder moves rg with depression from 0.353 to 0.420.
Pairing the two estimates block by block keeps the noise to 1.96 SE = 0.014, so p = 2.8e-20.